Two_bit_full_adder

Results Summary
The gate-level, structural, and behavioral implementations of the 2-bit full adder were verified against all 32 input combinations and produced identical sum and cout outputs in every case, confirming correct carry propagation and functional equivalence across all three design styles.

Transcript

```
# t=0 a1a0=00 b1b0=00 cin=0 || gate: sum1sum0=00 cout=0 || struc: sum1sum0=00 cout=0 || behav: sum1sum0=00 cout=0
# t=10 a1a0=00 b1b0=00 cin=1 || gate: sum1sum0=01 cout=0 || struc: sum1sum0=01 cout=0 || behav: sum1sum0=01 cout=0
# t=20 a1a0=00 b1b0=01 cin=0 || gate: sum1sum0=01 cout=0 || struc: sum1sum0=01 cout=0 || behav: sum1sum0=01 cout=0
# t=30 a1a0=00 b1b0=01 cin=1 || gate: sum1sum0=10 cout=0 || struc: sum1sum0=10 cout=0 || behav: sum1sum0=10 cout=0
# t=40 a1a0=00 b1b0=10 cin=0 || gate: sum1sum0=10 cout=0 || struc: sum1sum0=10 cout=0 || behav: sum1sum0=10 cout=0
# t=50 a1a0=00 b1b0=10 cin=1 || gate: sum1sum0=11 cout=0 || struc: sum1sum0=11 cout=0 || behav: sum1sum0=11 cout=0
# t=60 a1a0=00 b1b0=11 cin=0 || gate: sum1sum0=11 cout=0 || struc: sum1sum0=11 cout=0 || behav: sum1sum0=11 cout=0
# t=70 a1a0=00 b1b0=11 cin=1 || gate: sum1sum0=00 cout=1 || struc: sum1sum0=00 cout=1 || behav: sum1sum0=00 cout=1
# t=80 a1a0=01 b1b0=00 cin=0 || gate: sum1sum0=01 cout=0 || struc: sum1sum0=01 cout=0 || behav: sum1sum0=01 cout=0
# t=90 a1a0=01 b1b0=00 cin=1 || gate: sum1sum0=10 cout=0 || struc: sum1sum0=10 cout=0 || behav: sum1sum0=10 cout=0
# t=100 a1a0=01 b1b0=01 cin=0 || gate: sum1sum0=10 cout=0 || struc: sum1sum0=10 cout=0 || behav: sum1sum0=10 cout=0
# t=110 a1a0=01 b1b0=01 cin=1 || gate: sum1sum0=11 cout=0 || struc: sum1sum0=11 cout=0 || behav: sum1sum0=11 cout=0
# t=120 a1a0=01 b1b0=10 cin=0 || gate: sum1sum0=11 cout=0 || struc: sum1sum0=11 cout=0 || behav: sum1sum0=11 cout=0
# t=130 a1a0=01 b1b0=10 cin=1 || gate: sum1sum0=00 cout=1 || struc: sum1sum0=00 cout=1 || behav: sum1sum0=00 cout=1
# t=140 a1a0=01 b1b0=11 cin=0 || gate: sum1sum0=00 cout=1 || struc: sum1sum0=00 cout=1 || behav: sum1sum0=00 cout=1
# t=150 a1a0=01 b1b0=11 cin=1 || gate: sum1sum0=01 cout=1 || struc: sum1sum0=01 cout=1 || behav: sum1sum0=01 cout=1
# t=160 a1a0=10 b1b0=00 cin=0 || gate: sum1sum0=10 cout=0 || struc: sum1sum0=10 cout=0 || behav: sum1sum0=10 cout=0
# t=170 a1a0=10 b1b0=00 cin=1 || gate: sum1sum0=11 cout=0 || struc: sum1sum0=11 cout=0 || behav: sum1sum0=11 cout=0
# t=180 a1a0=10 b1b0=01 cin=0 || gate: sum1sum0=11 cout=0 || struc: sum1sum0=11 cout=0 || behav: sum1sum0=11 cout=0
# t=190 a1a0=10 b1b0=01 cin=1 || gate: sum1sum0=00 cout=1 || struc: sum1sum0=00 cout=1 || behav: sum1sum0=00 cout=1
```

Project : fa | Now: 320 ns Delta: 0 | sim:/two_bit_full_adder_tb/#INITIAL#37